



DANISH MIDTERM HANIF FILES FOR LMS HANDLING: 0323-1731413



CS304 Quiz File

السلام وعلیکم!

امید کرتا ہوں کہ آپ سب خیریت سے ہوں گے کافی دنوں کی محنت کے بعد الحمد للہ
ورچوئل یونیورسٹی کے مڈ ٹرم امتحان کے لیے فائلز تیار کر لی ہیں امید کرتا ہوں کہ آپ
سب سٹوڈنٹس کے امتحانات میں بہت فائدہ مند ہوں گی۔ فائلز میں کسی قسم کی کوئی بھی
غلطی ہو گئی ہو تو اس کے لیے معذرت خواہ ہیں انشاء اللہ مزید محنت سے ان فائلز کو بہتر بنائیں
گے۔ آپ سب سٹوڈنٹس سے گزارش ہے کہ ہمارے یوٹیوب چینل اور ویب سائٹ کو فالو
کر لیں۔

مکمل پیپرز آپ کو میری ویب سائٹ پہ ملیں گے لنک نیچے دیا گیا ہے۔

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ہیں تو اس نمبر پہ واٹس ایپ کریں۔

+923231731413

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Question #1: Constant data members of a class can be initialized by using _____.

- a) Constant Member Function
- b) Member Initializer List
- c) Class
- d) Non-constant Member Function

Correct Answer: b) Member Initializer List

Question #2: Consider the following code and choose the correct order of initialization for the data members of the class.

```
class A {  
    int a, b, c;  
  
public:  
    A(): b(10), c(0), a(7){}  
};
```

Select the correct option:

- a) a, c, b
- b) a, b, c
- c) b, a, c
- d) b, c, a

Correct Answer: b) a, b, c

Question #3: Which of the following can be used as an alternative to static members? Select the correct option:

- a) Constant Variables



- b) Global Variables
- c) Array of Objects
- d) Local Variables

Correct Answer: b) Global Variables

Question #4: In operator overloading, which of the following operators takes one or no arguments? Select the correct option:

- a) %
- b) <<
- c) *
- d) ++

Correct Answer: d) ++

Question #5: In C++, the subscript operator [] must be overloaded as a _____ of the class, with one parameter of _____ type. Select the correct option:

- a) Member function, char
- b) Member function, int
- c) Non-member function, char
- d) Non-member function, int

Correct Answer: b) Member function, int

Question #6: We can access a private static data member of a class through _____.

- a) Global member function
- b) Static member function
- c) Static data member
- d) Global data member

Correct Answer: b) Static member function

Question #7: If you do not initialize a static data member of a class of int type, it is automatically initialized with _____.

- a) 0
- b) Null
- c) 1
- d) -32767

Correct Answer: a) 0

Question #8: Memory is allocated to non-static members only when _____.

- a) Program is compiled
- b) Object is destructed
- c) Class is created
- d) Object is created

Correct Answer: d) Object is created

Question #9: In composition, _____ are called from composed objects to composing objects. Select the correct option:

- a) Destructors
- b) Variables
- c) Pointers
- d) Constructors

Correct Answer: a) Destructors

Question #10: `Student *ptr = new Student("Ali");` is an example of _____.

- a) Destructor
- b) Static memory allocation
- c) Dynamic memory allocation
- d) Constructor

Correct Answer: c) Dynamic memory allocation

Question #1: If class B inherits from class A, which of the following is the parent class?

- a) Both Class A & Class B
- b) Class A
- c) No parent class
- d) Class B

Answer: b) Class A

Question #2: How can the information hidden within an object be accessed?

- a) Through its private member functions
- b) Through both public and private members
- c) Through its private data members
- d) Through its interface

Answer: b) Through both public and private members

Question #3: How can we identify classes from a given problem statement?

- a) By extracting Nouns from the problem statement
- b) By extracting Verbs from the problem statement

- c) By extracting Adjectives from the problem statement
- d) By extracting Adverbs from the problem statement

Answer: a) By extracting Nouns from the problem statement

Question #4: A relationship in which one class object gets destroyed if the other class object is destroyed.

- a) Composition
- b) Association
- c) Inheritance
- d) Aggregation

Answer: a) Composition

Question

```
#include<iostream>
using namespace std;
int main() {
    int const x = 10;
    cout << ++x;
}
```

Options:

- 1. 11
- 2. Compile time Error
- 3. 10
- 4. Run Time Error

Answer: 2. Compile time Error

Question

```
class test {
    int a;

    int b;
```



```
int c;  
  
public:  
  
    test():b(5),c(a),a(b){}  
  
};
```

Options:

1. 5, 5, 5
2. Junk value, 5, 5
3. Junk value, 5, Junk value
4. 5, Junk value, 5

Answer: 3. Junk value, 5, Junk value

Question 3:

Which of the following is an example of a unary operator?

Options:

1. Inequality Operator (!=)
2. Increment Operator (++)
3. Equality Operator (==)
4. Division Operator (/)

Answer: 2. Increment Operator (++)

Question 4:

In the expression `b = c / d`, the arity of the division (/) operator is _____:

Options:

1. 2
2. 4
3. 1
4. 3

Answer:

1. 2

Question 5:

Static Data Member of a class is declared _____.

Options:

1. Outside the class
2. Inside main() method
3. Outside the main() method
4. Inside the class

Answer: 4. Inside the class

Question 6:

Which of the following statements is NOT true about a static variable of a class?

Options:

1. They remain in memory even when all objects of a class are destroyed.
2. Static variable belongs to a particular instance of a class.
3. They are created even when there is no object of a class.
4. Static variable is initialized once only throughout the program.

Answer: 2. Static variable belongs to a particular instance of a class.

Composition is _____ and aggregation is _____ relationship.

Options:

1. Weak, Strong
2. Strong, Strong
3. Weak, Weak
4. Strong, Weak

Answer:

1. **Weak, Strong**

Question 8:

Suppose we have a class named "Test" and `void show()` is one of its public member functions. Which of the following is a correct way to call `show()` function using a Test pointer `ptr`?

Options:

1. `ptr->show();`
2. `ptr::show();`
3. `ptr.show();`
4. `ptr.show();`

Answer:

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1. `ptr->show();`

Question 9:

In which of the following OOP paradigms, objects cannot exist independently?

Options:

1. Polymorphism
2. Inheritance
3. Composition
4. Aggregation

Answer: 3. **Composition**

Question 10:

Suppose we have a class named "Student"; which of the following statement is correctly declaring the pointer to the object for "Student" class?

Options:

1. `Student obj;`
2. `Student &obj;`
3. `*Student obj;`
4. `Student *obj;`

Answer: 4. `Student *obj;`

Question #5: Which of the following statement best describes the Constructor?

- a) Constructor is used to delete the objects of a class.
- b) Constructor is used to create an object of a class.
- c) Constructor is used to modify constant data members of the class.
- d) Constructor is used to initialize the data members of a class.

Answer: d) Constructor is used to initialize the data members of a class.

Question #6: All objects of the same class have _____ attributes.

- a) Same
- b) Integer
- c) Double
- d) Different

Answer: a) Same

Question #7: In OOP, which of the following feature is used to derive a class from another class?

- a) Polymorphism
- b) Data hiding
- c) Encapsulation
- d) Inheritance

Answer: d) Inheritance

Question #8: Which of the following can be the attribute of an object "Ali"?

- a) Eat
- b) Sleep
- c) Walk
- d) Address

Answer: d) Address

Question #9: A class can have only _____ destructor(s).

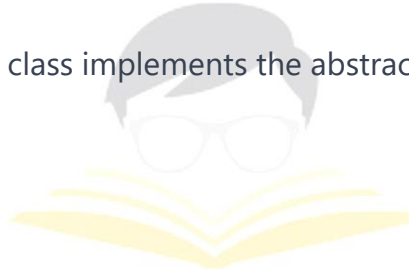
- a) Four
- b) Two
- c) One
- d) Three

Answer: c) One



Question #10: A/An _____ class implements the abstract concept.

- a) Public Class
- b) Concrete Class
- c) Friend Class
- d) Abstract Class



Answer: b) Concrete Class

Question #1: In OOP, we can achieve reusability through _____.

- A) Inheritance
- B) Association
- C) Data Hiding
- D) Encapsulation

Select the correct option.

Answer: A) Inheritance

Question #2: The concept of derived classes is involved in _____.

- A) Abstraction
- B) Encapsulation
- C) Inheritance
- D) Aggregation

Select the correct option.

Answer: C) Inheritance



Question #3: Minimum classes required in a program for implementing multiple inheritance:

- A) 3
- B) 2
- C) 5
- D) 4

Select the correct option.

Answer: B) 2

Question #4: Which of the following is an advantage of inheritance?

- A) Redundancy
- B) Ambiguity

C) Inconsistency

D) Re-usability

Select the correct option.

Answer: D) Re-usability

Question #5: Which of the following is NOT True about Destructor?

A) Used to free memory allocated through static allocation

B) Used to free memory allocated through dynamic allocation

C) Used to perform housekeeping operations

D) Free memory allocated using the new operator

Select the correct option.

Answer: A) Used to free memory allocated through static allocation

Question #6: Which of the following concept is more close to encapsulation?

A) Information Hiding

B) Inheritance

C) Polymorphism

D) Exception Handling

Select the correct option.

Answer: A) Information Hiding

Question #7: Identify the common attributes of the given scenario:

Circle has color, vertices, and radius. Triangle has color, vertices, and angle.

A) Vertices, Radius

B) Color, Angle

C) Color, Vertices

D) Vertices, Radius

Select the correct option.

Answer: C) Color, Vertices

Question #8: Which of the following notation represents the given statement? "Ali can drive more than one car."

A) 1 ... 1

B) 1* ... 1

C) 1 ... 0

D) 1 ... *

Select the correct option.

Answer: D) 1 ... *

Question #9: Member functions defined inside a class are _____ by default.

A) Private

B) Visible

C) Protected

D) Public



Select the correct option.

Answer: D) Public

Question #10: In a constant member function, the type of the this pointer is _____.

- A) Constant pointer to object
- B) Constant pointer to class
- C) Constant pointer
- D) Constant pointer to constant data

Select the correct option.

Answer: A) Constant pointer to object

Question #1: Which of the following is an example of a unary operator?

- A) Equality Operator (==)
- B) Division Operator (/)
- C) Increment Operator (++)
- D) Inequality Operator (!=)

Answer: C) Increment Operator (++)

Question #2: Constant data members of a class can be initialized by using _____.

- A) Member Initializer List
- B) Constant Member Function
- C) Class

D) Non-constant Member Function

Answer: A) Member Initializer List

Question #3: Which of the following can be created without any object of a class?

A) Instance Variables

B) Array of Objects

C) Non-Static Data Members

D) Static Data Members

Answer: D) Static Data Members

Question #4: Identify which of the following overloaded operator function's declaration is appropriate for the given call?

Rational_number_1 + 2.325

Where Rational_number_1 is an object of the user-defined class Rational_number.

A) Rational_number operator+(Rational_number &obj, double& num);

B) None of the given choices

C) operator+(double& obj);

D) Rational_number operator+(Rational_number & obj);

Answer: D) Rational_number operator+(Rational_number & obj);

Question #5: Which of the following can be used as an alternative to static members?

A) Constant Variables

B) Local Variables

C) Global Variables

D) Array of Objects

Answer: A) Constant Variables

Question #6: Choose the correct option to declare an overloaded stream insertion operator for a user-defined class "String" as a non-member friend function.

A) friend void operator << (const String & rhs);

B) friend void & operator << (ostream & os);

C) friend ostream & operator << ();

D) friend ostream & operator << (ostream & os, const String & s);

Answer: D) friend ostream & operator << (ostream & os, const String & s);

Question #7: What will be the value of variables a, b, and c after instantiating an object of the given class?

```
class test {  
    int a;  
    int b;  
    int c;  
public:  
    test(): b(5), c(a), a(b) {}  
};
```

A) 5, 5, 5

B) Junk value, 5, Junk value

C) Junk value, 5, 5

Answer: C) Junk value, 5, 5

Question #8: What is the general syntax of overloading Unary Operator as a member function of the class?

- A) TYPE & operator OP (TYPE &, TYPE &);
- B) TYPE & operator OP (TYPE &);
- C) TYPE & operator OP ();
- D) TYPE & operator OP (int);

Answer: B) TYPE & operator OP (TYPE &);

Question #9: What will be the output of the following code?

```
#include<iostream>
using namespace std;
int main() {
    int const x = 10;
    cout << ++x;
}
```

- A) 11
- B) 10
- C) Run Time Error
- D) Compile time Error

Answer: D) Compile time Error

Question #10: Composition is _____ and aggregation is _____ relationship.



A) Strong, Strong

B) Weak, Strong

C) Weak, Weak

D) Strong, Weak

Answer: D) Strong, Weak

